

18.4.1 The Expected Value of a Uniform Random Variable

Rolling a 6-sided die provides an example of a uniform random variable. Let R be the value that comes up when you roll a fair 6-sided die. Then by (18.2), the expected value of R is

$$\text{Ex}[R] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{7}{2}.$$

This calculation shows that the name "expected" value is a little misleading; the random variable might *never* actually take on that value. No one expects to roll a $3\frac{1}{2}$ on an ordinary die!

In general, if R_n is a random variable with a uniform distribution on $\{a_1, a_2, \dots, a_n\}$, then the expectation of R_n is simply the average of the a_i 's:

$$\text{Ex}[R_n] = \frac{a_1 + a_2 + \dots + a_n}{n}.$$

18.4.2 The Expected Value of a Reciprocal Random Variable

Define a random variable S to be the reciprocal of the value that comes up when you roll a fair 6-sided die. That is, $S = 1/R$ where R is the value that you roll.

define median (中位数),

随机变量小于该值的概率

$$\text{Ex}[S] = \text{Ex}\left[\frac{1}{R}\right] = \frac{1}{1} \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{1}{6} + \frac{1}{5} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{49}{120}.$$

Notice that

$$\text{Ex}[1/R] \neq 1/\text{Ex}[R].$$

Assuming that these two quantities are equal is a common mistake.

随机变量大于该值的概率

等于 1 减去 小于 1/2

18.4.3 The Expected Value of an Indicator Random Variable

The expected value of an indicator random variable for an event is just the probability of that event.

Lemma 18.4.2. If I_A is the indicator random variable for event A , then

$$\text{Ex}[I_A] = \text{Pr}[A].$$

⇒ eg. 丢硬币.

X=3: $\text{Pr}(<3) = 2/6$ (V) Proof.

$$\text{Pr}(>3) = 1/2$$

$$\text{Ex}[I_A] = 1 \cdot \text{Pr}[I_A = 1] + 0 \cdot \text{Pr}[I_A = 0] = \text{Pr}[I_A = 1]$$

$$= \text{Pr}[A]. \quad (\text{def of } I_A)$$

✓ 4: $\text{Pr}(<4) = 1/2$ (V)

For example, if A is the event that a coin with bias p comes up heads, then $\text{Ex}[I_A] = \text{Pr}[I_A = 1] = p$.

$$\text{Pr}(>4) = 2/6$$